

IE-20: Exact Scalar Threshold and Extended Precision Bounds

Proofs, exact checks, and the remaining matching problem

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Completion status. This is an *extended partial resolution*, not a full solution to IE-20. It adds an exact scalar answer, a sharper one-step result, a joint dimension/conditioning breakdown family, a capped-condition-number upper bound, and two finite-precision early-convergence arguments. The upper and lower bounds still do not match for all parameter triples. Section 9 gives a concrete remaining gap. The proofs are not independently refereed or proof-assistant verified.

Abstract. We study the fixed CG recurrence and adversarial scalar relative-error envelope specified by IE-20. The scalar threshold is determined exactly by a degree-five rational inequality. A prefix-wise analysis sharpens the dimension factor in the local matvec error. In general dimension, set

$$H = \min\{K, 64/\varepsilon^2\}, \quad D = 1024H^2,$$

and let m be the minimum of the dimension, an explicit Chebyshev horizon of order $\sqrt{K} \log(K/\varepsilon)$, and a backward-error horizon of order $\varepsilon^{-2/3}$. We prove the sufficient precision

$$\left\lceil \log_2 \frac{1024(n+1)KD^{12m+20}}{\varepsilon^2} \right\rceil.$$

The cap in H follows from the fact that an excessively long step would already have small backward error. The early-horizon arguments transfer polynomial approximation to the actual rounded recurrence, rather than assuming that computed iterates are exact Krylov projections. A prefix-wise matvec estimate and a new breakdown family also give

$$P_{\text{CG}}(n, K, \varepsilon) = \log_2(nK) + O_\varepsilon(1)$$

for each fixed tolerance, uniformly in $n \geq 2$ and $K \geq 1$. The additive constants are universal when $3/8 \leq \varepsilon < 1/2$. There is also a sharp order uniformly for $1 \leq K \leq 2$. The previous path/outlier lower bound and its exact certificates are preserved in the companion archive.

The package contains this PDF, editable L^AT_EX, standard-library Python code, regenerated exact checks, a proof-scope audit, source notes, and the previous recovery archive. Finite checks verify their recorded examples, not the universal analytic claims.

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1 Problem, notation, and result map

We use the editorial IE-20 formulation in the repository[1]. For $n \geq 1$, $K \geq 1$, and $0 < \varepsilon < 1/2$, stored inputs are SPD $A \in \mathbb{R}^{n \times n}$ and $b \neq 0$, with $\kappa_2(A) \leq K$ and at most p binary significant bits per entry. Exponents are unbounded. Each scalar operation satisfies

$$\text{fl}(a \circ b) = (a \circ b)(1 + \delta), \quad |\delta| \leq u = 2^{-p}.$$

Errors are independently admissible. Products and additions are separate; dot products include all indices, accumulated from zero in increasing order. Copies and comparisons are exact. The recurrence stops at a zero recursive residual or before a zero divisor. Success requires a produced iterate of index at most n satisfying

$$\eta(x) = \frac{\|b - Ax\|_2}{\|A\|_2 \|x\|_2 + \|b\|_2} \leq \varepsilon. \quad (1)$$

The threshold $P_{\text{CG}}(n, K, \varepsilon)$ must guarantee success at every greater precision too, for every admissible input and error schedule.

Interpretation of the error model. The error equations are an envelope, not a correctly rounded binary map. An addition of zero may carry an allowed relative error, and an exact intermediate may be retained even if it is not a p -bit number. Input representability is a separate condition. All lower-bound constructions and the exact executor use this interpretation. A lower bound here is not automatically a lower bound for IEEE round-to-nearest.

We write a search direction as v_j , reserving p for precision:

$$\begin{aligned} x_0 &= 0, \quad r_0 = v_0 = b, \quad \rho_0 = \text{fl}(b^T b), \\ q_j &= \text{fl}(Av_j), \quad d_j = \text{fl}(v_j^T q_j), \quad \alpha_j = \text{fl}(\rho_j/d_j), \\ x_{j+1} &= \text{fl}(x_j + \alpha_j v_j), \quad r_{j+1} = \text{fl}(r_j - \alpha_j q_j), \\ \rho_{j+1} &= \text{fl}(r_{j+1}^T r_{j+1}), \quad \beta_j = \text{fl}(\rho_{j+1}/\rho_j), \quad v_{j+1} = \text{fl}(r_{j+1} + \beta_j v_j). \end{aligned}$$

The last line is evaluated only when another step is needed. Every product and addition inside a displayed vector expression is separately rounded according to the preceding conventions.

1.1 The strengthened upper bound

Define the integer horizons

$$\bar{K} = \max\{2, K\}, \quad k_C = \left\lceil \sqrt{\bar{K}} \left\lceil \log_2 \frac{16\bar{K}}{\varepsilon} \right\rceil \right\rceil, \quad (2)$$

$$k_U = \left\lceil (64/\varepsilon)^{2/3} \right\rceil, \quad m = \min\{n, k_C, k_U\}, \quad (3)$$

$$H = \min\{K, 64/\varepsilon^2\}, \quad D = 1024H^2. \quad (4)$$

The main upper-bound theorem, proved in Sections 5–7, is

$$P_{\text{CG}}(n, K, \varepsilon) \leq \max \left\{ 2, \left\lceil \log_2 \frac{1024(n+1)KD^{12m+20}}{\varepsilon^2} \right\rceil \right\}. \quad (5)$$

Consequently, with universal constants,

$$P_{\text{CG}} = O \left(\log \frac{nK}{\varepsilon} + \min \left\{ n, \sqrt{\bar{K}} \log \frac{K}{\varepsilon}, \varepsilon^{-2/3} \right\} \log \left(2 \min\{K, \varepsilon^{-2}\} \right) \right). \quad (6)$$

The constants in (5) are conservative. The integer horizon k_C is deliberately larger than the least degree satisfying the Chebyshev estimate.

1.2 What is sharp, and what is not

For $n = 1$, Section 2 gives the exact integer threshold. For $n \geq 2$, the general lower bound remains

$$P_{\text{CG}}(n, K, \varepsilon) \geq c \left(\log_2 \frac{n}{\varepsilon} + \log_2 K \right). \quad (7)$$

Combining the new upper arguments with explicit witnesses yields

$$P_{\text{CG}}(n, K, \varepsilon) = \Theta(\log_2(n/\varepsilon)), \quad 1 \leq K \leq 2, \quad (8)$$

$$P_{\text{CG}}(n, K, \varepsilon) = \Theta(\log_2(nK)), \quad n \geq 2, \quad 3/8 \leq \varepsilon < 1/2, \quad (9)$$

$$P_{\text{CG}}(2, K, \varepsilon) = \Theta(\log_2 K + \log_2(1/\varepsilon)). \quad (10)$$

All constants in these displayed Θ statements are universal on the indicated domains. For each fixed tolerance, uniformly in $n \geq 2$ and $K \geq 1$, we prove

$$P_{\text{CG}}(n, K, \varepsilon) = \log_2(nK) + O_\varepsilon(1). \quad (11)$$

In particular, for fixed $n \geq 2$ and fixed ε ,

$$P_{\text{CG}}(n, K, \varepsilon) = \log_2 K + O_{n,\varepsilon}(1). \quad (12)$$

None of these statements closes the all-parameter gap between (7) and (6).

1.3 Relation to the recovered work and recent sources

The prior manuscript in `prior/IE20_recovered.zip` supplied the same elementary lower bounds, the $K = 1$ and $n = 2$ orders, an upper bound with $H = K$, $m = n$, and an extra dimension factor in the local error budget, and a path/outlier family requiring order $n \log K$ on specified joint parameter sequences. This manuscript proves the new upper statements directly, while retaining the old archive for the full hard-family proof and provenance. No claim of external priority is made.

Dereziński, Nakatsukasa, and Rebrova give an exact-arithmetic reciprocal polynomial approximation used in Case 3 of Theorem 7.1[2]. The transfer to rounded CG iterates is proved here; it is not imported as a finite-precision conclusion from their theorem.

Bake, Carson, and Ma establish error estimates at a sufficiently late iteration under stated smallness conditions; their theorem does not bound that iteration by n [3]. A September 2026 manuscript by Sinha gives a sufficient precision for a specified CG computation with independently recomputed residual tests, together with a conditioning obstruction, and explicitly retains a quantitative gap[4]. These results do not supply the missing matching characterization for the present target.

2 An exact answer in dimension one

Theorem 2.1 (Exact scalar threshold). *For every $K \geq 1$, define*

$$E_1(u) = \frac{(1+u)^4 - (1-u)^5}{(1+u)^4 + (1-u)^5}. \quad (13)$$

Then

$$P_{\text{CG}}(1, K, \varepsilon) = \min\{p \in \mathbb{Z} : p \geq 2, E_1(2^{-p}) \leq \varepsilon\}. \quad (14)$$

In particular, the answer is independent of the supplied condition-number bound K and is computable by exact rational comparisons.

Proof. Write $A = [a]$, $a > 0$, and $b \neq 0$. Exactly nine elementary operations contribute to the first iterate: two for ρ_0 , two for q_0 , two for d_0 , one for division, and two for the update of x_1 . All relevant divisors are positive because $u \leq 1/4$. After cancellation of the input values,

$$x_1 = \frac{b}{a}F,$$

where F is a product of five allowed factors $1 + \delta$ divided by four such factors. Thus

$$F_- := \frac{(1-u)^5}{(1+u)^4} \leq F \leq F_+ := \frac{(1+u)^5}{(1-u)^4}.$$

For positive F , the backward error is $|1 - F|/(1 + F)$. It increases with $|\log F|$, and

$$-\log F_- - \log F_+ = -\log(1 - u^2) > 0.$$

The worse endpoint is therefore F_- . Its error is precisely (13). This endpoint is attained with the admissible inputs $a = b = 1$, errors $-u$ in the five numerator operations and $+u$ in the four denominator operations. The initial iterate has error one. Residual-update operations occur after x_1 has been produced and cannot alter its backward error.

The function $F_-(u)$ is strictly decreasing on $(0, 1/4]$, so $E_1(u)$ is strictly increasing. Consequently the least successful precision also guarantees success at every larger precision. This proves the threshold quantifier in (14). $\square$

The leading small- u behavior is $E_1(u) = (9/2)u + O(u^2)$, so $P_{\text{CG}}(1, K, \varepsilon) = \log_2(1/\varepsilon) + O(1)$. The exact formula, rather than this expansion, handles equality cases and integer rounding. The companion code exhausts all 2^9 sign corners at each of three precisions; the analytic monotonicity argument covers the entire continuous error box.

3 Elementary lower bounds retained for comparison

Theorem 3.1. *For every admissible triple,*

$$P_{\text{CG}}(n, K, \varepsilon) \geq \max \left\{ 2, 1 + \left\lfloor \log_2 \frac{n}{4\varepsilon} \right\rfloor \right\}. \quad (15)$$

For $n \geq 2$ there is also the bound

$$P_{\text{CG}}(n, K, \varepsilon) \geq \max\{2, \lfloor \log_2(K+1) \rfloor\}. \quad (16)$$

Together they imply (7).

Proof. For (15), choose $A = I_n$ and $b = e_1$. Set the error to $+u$ on every addition in the first row of the first matrix-vector product, including additions of zero, and set every other error to zero. Then $q_0 = te_1$ with $t = (1+u)^n$. The produced iterate and recursive residual are $x_1 = t^{-1}e_1$ and $r_1 = 0$. The algorithm stops, but its true backward error is $(t-1)/(t+1)$. For $u \leq 1/4$ and $\varepsilon < 1/2$,

$$\log(1+u) \geq \frac{4}{5}u, \quad \log \frac{1+\varepsilon}{1-\varepsilon} \leq \frac{8}{3}\varepsilon.$$

Hence $u \geq 4\varepsilon/n$ gives strict failure. Failure at $p = \lfloor \log_2(n/(4\varepsilon)) \rfloor \geq 2$ forces the threshold to exceed p . Smaller values are covered by its defining lower bound two.

For (16), at precision $p \geq 2$ set $u = 2^{-p}$ and

$$A = \begin{pmatrix} 1 & 1-u \\ 1-u & 1 \end{pmatrix} \oplus I_{n-2}, \quad b = (1, -1, 0, \dots, 0)^T.$$

The stored number $1-u$ has exactly p significant bits. The condition number is $(2-u)/u = 2^{p+1}-1$. Put error $-u$ on the two diagonal products in the first matrix-vector product and zero elsewhere. Both nonzero rows now sum to zero, so $q_0 = d_0 = 0$. Only x_0 is produced, and it fails. If $q = \lfloor \log_2(K+1) \rfloor \geq 3$, take $p = q - 1$; then this input has condition number at most K and the threshold is at least q . The remaining cases again follow from $P_{\text{CG}} \geq 2$.

Finally, the maximum of the two logarithmic lower bounds is at least half their sum, up to additive constants absorbed by $P_{\text{CG}} \geq 2$. $\square$

Theorem 3.2 (Joint dimension/conditioning breakdown). *Let $n \geq 2$, $K \geq 15$, and $a = \lfloor n/2 \rfloor$. Then*

$$P_{\text{CG}}(n, K, \varepsilon) \geq 1 + \left\lfloor \log_2 \frac{a(K+1)}{2} \right\rfloor. \quad (17)$$

For all $n \geq 2$, $K \geq 1$, and admissible tolerances, this implies

$$P_{\text{CG}}(n, K, \varepsilon) \geq \log_2(nK) - 5. \quad (18)$$

Proof. Set $p = \lfloor \log_2(a(K+1)/2) \rfloor$, $u = 2^{-p}$, $t = au$, and $c = 1 - t$. These choices give

$$p \geq 2, \quad \frac{2}{K+1} \leq t < \frac{4}{K+1} \leq \frac{1}{4}, \quad nu < 1.$$

For the last estimate, use $u < 4/[a(K+1)]$ and $a \geq n/4$. Put the block $\begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix}$ on indices 1 and n , put identity on the remaining indices, and choose $b = e_1 - e_n$. The condition number is $(2-t)/t \leq K$. The stored entry $c = (2^p - a)/2^p$ has at most p significant bits.

In each active row, the first nonzero product is followed by $n-1$ additions before the final column is included. Thus one product and $n-1$ additions supply n independently selectable factors. In row 1, scale its initial value one down to c ; in row n , scale its initial value c up to one. Both scalings are admissible. Indeed,

$$t = au \leq \frac{nu}{2} \leq \frac{nu}{1+nu}, \quad c(1+u)^n \geq c(1+nu) \geq 1.$$

Thus $c \geq (1+u)^{-n} \geq (1-u)^n$ as well. For an exact rational schedule, repeatedly use factor $1-u$ on the first row until the next such factor would undershoot c , then use the exact factor needed to reach c and subsequently factor one. On the last row use the corresponding greedy factors up to $1+u$ toward one. Every chosen factor is rational and lies in the permitted interval.

Evaluate the last-column products and additions exactly. They cancel $c - c$ in row 1 and $1 - 1$ in row n . All other rows are zero, so $q_0 = d_0 = 0$. The run breaks down with only x_0 produced. Failure at precision p proves (17). Since $a \geq n/4$, this lower bound exceeds $\log_2(n(K+1)/8)$. For $K < 15$, the identity witness and $\varepsilon < 1/2$ give $P_{\text{CG}} > \log_2(n/2) > \log_2(nK) - 5$. Combining the two ranges proves (18). Together with the identity witness this also gives

$$P_{\text{CG}}(n, K, \varepsilon) \geq \log_2 \left(n \max\{K, \varepsilon^{-1}\} \right) - 5.$$

$\square$

4 The sharp exact first-step envelope

Theorem 4.1. *For exact arithmetic on real SPD inputs with $n \geq 2$ and condition number at most K , the largest possible first-step backward error is*

$$M(K) = \frac{K-1}{\sqrt{8K(K+1)}}. \quad (19)$$

This is a real-input extremal statement; exact representability of its maximizing vector at a prescribed finite precision is not asserted.

Proof. Normalize $\|A\|_2 = \|b\|_2 = 1$, put $a = 1/K$, and write $\mu = b^T A b$, $\nu = b^T A^2 b$. The exact first iterate is $x = b/\mu$, so

$$\eta(x)^2 = \frac{\nu - \mu^2}{(1 + \mu)^2}.$$

Every eigenvalue $t \in [a, 1]$ satisfies $t^2 \leq (1 + a)t - a$. Consequently $\nu - \mu^2 \leq (1 - \mu)(\mu - a)$. The identity

$$\frac{(1 - a)^2}{8(1 + a)} - \frac{(1 - t)(t - a)}{(1 + t)^2} = \frac{((a + 3)t - (3a + 1))^2}{8(1 + a)(1 + t)^2}$$

shows that the maximum occurs at $\mu = (1 + 3a)/(3 + a)$. The resulting square root is (19). For $K > 1$, equality is attained by a two-point spectrum $\{a, 1\}$ and a unit vector whose squared weights are $2/(3 + a)$ at a and $(1 + a)/(3 + a)$ at one. Extra coordinates may have zero right-hand-side weight. For $K = 1$, the error is identically zero. $\square$

Theorem 4.2 (Finite-precision first step). *If $\varepsilon > M(K)$, then*

$$P_{\text{CG}}(n, K, \varepsilon) \leq \max \left\{ 2, \left\lceil \log_2 \frac{512(n + 1)K}{\varepsilon - M(K)} \right\rceil \right\}. \quad (20)$$

The conclusion also holds for $n = 1$, although Theorem 2.1 is sharper there.

Proof. Continue with $\|A\|_2 = \|b\|_2 = 1$. Set $\omega = 64(n + 1)Ku$ and suppose $\omega \leq 1/16$. The local scalar estimates proved in Lemma 5.2 give $\alpha_0\mu = 1 + \theta$, $|\theta| \leq \omega$, and ensure a positive computed denominator. The separately rounded product and addition in x_1 then give

$$\|x_1 - b/\mu\|_2 \leq \frac{2\omega}{\mu}.$$

Indeed their additional relative deviation is at most $2u + u^2$, and $u \leq \omega/128$. Perturbing the numerator and denominator of (1), with $x^{\text{ex}} = b/\mu$, yields

$$\eta(x_1) \leq \frac{\eta(x^{\text{ex}}) + 2\omega}{1 - 2\omega} \leq M(K) + 4\omega.$$

For the last inequality, use $M(K) < 1/(2\sqrt{2})$ and $2\omega \leq 1/8$. Let $\Delta = \varepsilon - M(K) > 0$. The precision in (20) implies $\omega \leq \Delta/8 < 1/16$, so $\eta(x_1) \leq M(K) + \Delta/2 < \varepsilon$. This holds at all larger precisions as well. $\square$

Corollary 4.3. *For $n \geq 2$ and $3/8 \leq \varepsilon < 1/2$, (9) holds. Also, whenever $n \geq 2$ and $\varepsilon \geq 2M(K)$,*

$$P_{\text{CG}}(n, K, \varepsilon) = \Theta\left(\log_2 \frac{nK}{\varepsilon}\right)$$

with universal constants on this region.

Proof. Since $1/(2\sqrt{2}) < 23/64$, the coarse-tolerance region has $\varepsilon - M(K) \geq 1/64$. Equation (20) is then at most $\lceil \log_2(32768(n + 1)K) \rceil$. The joint lower bound below improves this to the explicit additive comparison

$$\log_2(nK) - 5 \leq P_{\text{CG}}(n, K, \varepsilon) \leq \log_2(nK) + 17.$$

In the second region, $\varepsilon - M(K) \geq \varepsilon/2$, and the same comparison proves the stated order. $\square$

5 A failure-conditioned bootstrap

This section supplies the common recurrence analysis for all three horizons in (3). The important new point is that H in (4), rather than K , controls the step lengths while every produced iterate fails.

5.1 Normalization and local error equations

Homogeneity allows $\|A\|_2 = \|b\|_2 = 1$. To see this without assuming representability of normalized inputs, let $a_0 = \|A\|_2$ and $B_0 = \|b\|_2$ and transform an execution by

$$A' = A/a_0, \quad b' = b/B_0, \quad x'_j = a_0 x_j/B_0, \quad r'_j = r_j/B_0, \quad v'_j = v_j/B_0.$$

Then $\alpha'_j = a_0 \alpha_j$, $\rho'_j = \rho_j/B_0^2$, and β_j is unchanged. Every scalar error equation transforms with the same δ . The normalized analysis applies to all real inputs, hence also to this possibly nonrepresentable normalized input. The backward error is invariant. The normalized spectrum lies in $[1/K, 1]$.

Set

$$\omega = 64(n+1)Ku, \tag{21}$$

and write

$$R_j = \|r_j\|_2, \quad V_j = \|v_j\|_2, \quad X_j = \|x_j\|_2, \quad Q_j = v_j^T A v_j.$$

The symbol Q_j here is a scalar quadratic form, not a Lanczos basis.

Lemma 5.1 (Normwise prefix estimate). *For the specified left-to-right matvec, if $nu \leq 1/2$, then*

$$\|\mathfrak{fl}(Av) - Av\|_2 \leq 4nu \|A\|_2 \|v\|_2. \tag{22}$$

The estimate does not replace A by its entrywise absolute value.

Proof. Collect the row accumulators at their common column index j into a vector s_j . This is an algebraic grouping of a completed execution; it does not change the row-by-row evaluation order or its errors. Let $A_{:,j}$ be column j and let $y_j = \sum_{i=1}^j A_{:,i} v_i$ be its exact prefix. Diagonal matrices $D_j^{\text{add}}, D_j^{\text{mul}}$ of norm at most u encode the componentwise errors. For $e_j = s_j - y_j$,

$$e_j = D_j^{\text{add}} y_j + (I + D_j^{\text{add}})(e_{j-1} + D_j^{\text{mul}} A_{:,j} v_j).$$

Unrolling from $e_0 = 0$ gives

$$\|e_n\|_2 \leq (1+u)^n u \left(\sum_{j=1}^n \|y_j\|_2 + \sum_{j=1}^n \|A_{:,j}\|_2 |v_j| \right).$$

Each exact prefix has norm at most $\|A\|_2 \|v\|_2$, while the second sum is at most $\|A\|_F \|v\|_2 \leq \sqrt{n} \|A\|_2 \|v\|_2$. Since $(1+u)^n \leq e^{nu} < 2$, the result follows from $2(n + \sqrt{n}) \leq 4n$. All zero terms remain included. $\square$

Lemma 5.2 (Local envelopes). *Suppose $\omega \leq 1/2$. Whenever the relevant nonzero residuals and directions are available, their computed scalar divisors are positive, and the performed operations satisfy*

$$\alpha_j Q_j = R_j^2(1 + \theta_j), \quad |\theta_j| \leq \omega, \tag{23}$$

$$\beta_j R_j^2 = R_{j+1}^2(1 + \zeta_j), \quad |\zeta_j| \leq \omega, \tag{24}$$

$$r_{j+1} = r_j - \alpha_j A v_j + f_j, \quad \|f_j\|_2 \leq \omega(R_j + \alpha_j V_j), \tag{25}$$

$$x_{j+1} = x_j + \alpha_j v_j + g_j, \quad \|g_j\|_2 \leq \omega(X_j + \alpha_j V_j), \tag{26}$$

$$v_{j+1} = r_{j+1} + \beta_j v_j + h_j, \quad \|h_j\|_2 \leq \omega(R_{j+1} + \beta_j V_j). \tag{27}$$

Equations involving the next direction are used only when it is formed.

Proof. Let $\gamma = (n+1)u/[1 - (n+1)u]$. Expanding at most $n+1$ factors $1 + \delta$ in each dot-product term gives $\gamma \leq 2(n+1)u$. The assumption $\omega \leq 1/2$ implies $(n+1)Ku \leq 1/128$, so Lemma 5.1 gives

$$\|q_j - Av_j\|_2 \leq 4nuV_j.$$

The subsequent dot product satisfies

$$\begin{aligned} |d_j - Q_j| &\leq 4nuV_j^2 + 2(n+1)uV_j \|q_j\|_2 \\ &\leq 8(n+1)uV_j^2 \leq 8(n+1)KuQ_j. \end{aligned}$$

Its relative error is at most $\omega/8 \leq 1/16$, so $d_j > 0$ for $v_j \neq 0$. Also $|\rho_j - R_j^2| \leq \gamma R_j^2$, with relative error at most $\omega/32$, and $\rho_j > 0$ for $r_j \neq 0$. The rounded division has error at most $u \leq \omega/128$. Taking the two ratios therefore proves (23) and (24), with bounds strictly smaller than ω . A separately rounded product and vector addition contribute at most $3u$ times the sum of the input norms. Combining this observation with the matvec estimate proves (25)–(27). All estimates retain the additions of zero required by the specification. $\square$

5.2 Induction statement

Fix an integer $1 \leq m \leq n$, and use H, D from (4). Assume

$$\sigma^2 := \omega D^{12m+20} \leq \varepsilon^2/16. \quad (28)$$

Let E_j be the maximum absolute off-diagonal inner product among $r_0, \dots, r_j$, and the maximum absolute off-diagonal A -inner product among formed directions through v_j . At the endpoint $j = m$, include only directions through v_{m-1} ; v_m is not needed. Empty maxima are zero. Put $B = 32H$ and

$$w_j = b - Ax_j - r_j, \quad W_j = \|w_j\|_2.$$

Lemma 5.3 (Bootstrap under total failure). *Assume that every iterate produced through the horizon fails (1). Under (28), the execution cannot stop before producing x_m . For $0 \leq j \leq m$, it satisfies*

$$\max\{1, R_j, X_j, V_j\} \leq D^{j+1}, \quad (29)$$

$$E_j \leq \omega D^{8j+8}, \quad (30)$$

$$W_j \leq 4\omega D^{j+2} \leq \varepsilon/8, \quad (31)$$

$$R_j \geq \varepsilon X_j + 7\varepsilon/8. \quad (32)$$

The terms involving V_j concern only $j < m$. Every needed coefficient obeys

$$\frac{1}{4} \leq \alpha_j \leq 8H, \quad 0 \leq \beta_j \leq B, \quad (33)$$

and the corresponding local errors obey

$$\|f_j\|_2, \|g_j\|_2 \leq \omega D^{j+2}, \quad \|h_j\|_2 \leq 2\omega D^{j+2}. \quad (34)$$

Proof. The initial conditions have $R_0 = V_0 = 1$, $X_0 = W_0 = E_0 = 0$. At each subsequent index, (31) and failure imply

$$R_j \geq \|b - Ax_j\|_2 - W_j > \varepsilon(X_j + 1) - W_j \geq \varepsilon X_j + 7\varepsilon/8.$$

Thus $R_j > \sigma$. We establish the remaining estimates in the order below; this order is important for excluding breakdown without assuming that all steps already exist.

The current direction is nonzero. Expanding (27) gives a combination of residuals through r_j with coefficient sum at most $2B^j$, plus accumulated local errors. Since $B/D \leq 1/32$, their total norm is at most $3\omega D^{j+1}$. Consequently

$$\begin{aligned} |r_j^T v_j - R_j^2| &\leq 2D^j E_j + 3\omega D^{2j+2} \\ &\leq \omega D^{9j+10} \leq \sigma^2/8. \end{aligned}$$

For $j < m$, the last comparison follows from (28) and $D \geq 1024$. Since $R_j > \sigma$, it gives $V_j \geq 7R_j/8 > 0$. Lemma 5.2 now ensures $d_j > 0$, so the next step can legally be evaluated.

The upper bound on the step length. If $H = K$, use $Q_j \geq V_j^2/K$ in (23) to obtain $\alpha_j \leq 8H$. Suppose instead that $H = 64/\varepsilon^2$. Let $\alpha = \alpha_j$, $s = \alpha v_j$, and $S = \|s\|_2$. The induction gives $X_j \leq R_j/\varepsilon$, $W_j \leq R_j/7$, and $S \geq 7\alpha R_j/8$. If $\alpha > 8H = 512/\varepsilon^2$, these inequalities imply $X_j/S \leq 1/4$. Also $A^2 \preceq A$ and (23) give

$$\|As\|_2 \leq \sqrt{2\alpha} R_j, \quad \|g_j\|_2 \leq \omega(X_j + S).$$

The denominator in the next backward error is at least $S/2$. Using $b - Ax_j = r_j + w_j$ in its numerator therefore yields the safe bound

$$\eta(x_{j+1}) \leq \frac{5}{\alpha} + \frac{6}{\sqrt{\alpha}} + 3\omega < \varepsilon. \quad (35)$$

The last inequality follows from $\alpha > 512/\varepsilon^2$, $\varepsilon < 1/2$, and $\omega \leq \varepsilon/32$, all consequences of our assumptions. This contradicts failure of the newly produced iterate. Hence $\alpha_j \leq 8H$ in the capped case too. The argument has not presumed bounds on f_j, g_j beyond the local envelopes, nor presumed existence of a later coefficient.

The lower bound on the step length. For $j \geq 1$, write $v_j = r_j + \beta_{j-1}v_{j-1} + h_{j-1}$ and expand the quadratic form in the opposite direction. With $C = v_{j-1}^T A v_j$, this gives

$$\begin{aligned} Q_j &= r_j^T A r_j + 2\beta_{j-1}C - \beta_{j-1}^2 v_{j-1}^T A v_{j-1} \\ &\quad + 2h_{j-1}^T A(v_j - \beta_{j-1}v_{j-1}) - h_{j-1}^T A h_{j-1} \\ &\leq R_j^2 + 2BE_j + 2\|h_{j-1}\|_2 (V_j + BV_{j-1}) \\ &\leq R_j^2 + \omega D^{8j+10} \leq R_j^2 + \sigma^2/8 \leq 2R_j^2. \end{aligned}$$

For $j = 0$, $Q_0 \leq R_0^2$ directly. Equation (23) therefore gives $\alpha_j \geq 1/4$.

Next residual and iterate. The two update envelopes imply

$$R_{j+1}, X_{j+1} \leq (1 + \omega)(1 + 8H)D^{j+1} \leq \frac{1}{4}D^{j+2}, \quad \|f_j\|_2, \|g_j\|_2 \leq \omega D^{j+2}.$$

Their true-residual gap satisfies $w_{j+1} = w_j - Ag_j - f_j$. Summing gives (31) at the next index. If the new recursive residual were zero, this gap bound would already contradict failure. Thus a needed next norm-square divisor is positive.

Bounding the next ratio. Using $A^2 \preceq A$ and $r_j^T A v_j = Q_j - \beta_{j-1}v_{j-1}^T A v_j - h_{j-1}^T A v_j$ in the squared residual update gives

$$\begin{aligned} R_{j+1}^2 &\leq R_j^2 + (\alpha_j - 2)R_j^2(1 + \theta_j) + 2\alpha_j B E_j + 2\alpha_j \|h_{j-1}\|_2 V_j \\ &\quad + 2\|f_j\|_2 (R_j + \alpha_j V_j) + \|f_j\|_2^2 \\ &\leq 12HR_j^2 + DE_j + \omega D^{2j+4} \\ &\leq 12HR_j^2 + \omega D^{8j+10} \leq 13HR_j^2. \end{aligned}$$

At $j = 0$ the terms with index -1 are absent. For explicit constant checks, $2\alpha_j B \leq D/2$, while the other local terms are bounded by $235H^2\omega D^{2j+2}$, which is less than ωD^{2j+4} . Equation (24) yields $\beta_j \leq 32H = B$. Then

$$V_{j+1} \leq (1 + \omega)(R_{j+1} + BV_j) \leq \frac{3}{2}(\frac{1}{4} + \frac{1}{32})D^{j+2} < D^{j+2}.$$

The bound on h_j follows immediately.

Residual orthogonality defects. Put $F_j = \max\{E_j, |r_i^T r_{j+1}| : 0 \leq i \leq j\}$. For $i < j$, substitute $r_i = v_i - \beta_{i-1}v_{i-1} - h_{i-1}$ into $r_i^T r_{j+1} = r_i^T r_j - \alpha_j r_i^T A v_j + r_i^T f_j$. For $i = j$, use

$$r_j^T r_{j+1} = R_j^2 - \alpha_j Q_j + \alpha_j \beta_{j-1} v_{j-1}^T A v_j + \alpha_j h_{j-1}^T A v_j + r_j^T f_j.$$

In each case the coefficient of E_j is below D , and the remaining terms are bounded by ωD^{2j+5} . Hence

$$F_j \leq D E_j + \omega D^{2j+5}. \quad (36)$$

Direction conjugacy defects. When v_{j+1} is needed, use $A v_i = (r_i - r_{i+1} + f_i)/\alpha_i$. For $i < j$, this gives

$$|v_i^T A v_{j+1}| \leq 8F_j + B E_j + 2\omega D^{2j+3}.$$

For the adjacent pair the exact algebraic cancellation is

$$\begin{aligned} \alpha_j v_j^T A v_{j+1} &= r_j^T r_{j+1} - R_{j+1}^2 + f_j^T r_{j+1} + \alpha_j \beta_j Q_j + \alpha_j v_j^T A h_j, \\ \alpha_j \beta_j Q_j &= R_{j+1}^2 (1 + \theta_j)(1 + \zeta_j). \end{aligned}$$

Therefore $|v_j^T A v_{j+1}| \leq 4F_j + 2\omega D^{2j+4}$. Combining with (36) proves

$$E_{j+1} \leq D^2 E_j + \omega D^{2j+7} \leq \omega(D^{8j+10} + D^{2j+7}) \leq \omega D^{8(j+1)+8}.$$

At the horizon no next direction is required, so (36) alone suffices. This completes the induction and the exclusion of every premature stopping mechanism under the failure assumption. $\square$

6 Transfer from polynomials to computed spans

For a polynomial $\pi(t) = \sum c_i t^i$, let $\|\pi\|_{\text{coef},1} = \sum |c_i|$. The next lemma is an explicit substitute for an unjustified claim that rounded CG spans an exact Krylov space.

Lemma 6.1 (Approximate polynomial containment). *Under Lemma 5.3, fix $1 \leq k \leq m$. If $\deg \pi < k$ and $\|\pi\|_{\text{coef},1} \leq D^{k+1}$, then $y = \pi(A)b$ has a representation*

$$y = \sum_{i=0}^{k-1} c_i v_i + e, \quad \sum |c_i| \leq D^{2k}, \quad \|e\|_2 \leq \omega D^{3k+1}. \quad (37)$$

There is also a vector $z \in \text{span}\{x_1, \dots, x_k\}$ such that

$$\|z - y\|_2 \leq \omega D^{3k+2}. \quad (38)$$

Furthermore, for $i < k$,

$$|r_k^T v_i| \leq \omega D^{9k+10}. \quad (39)$$

Proof. For $i \leq k - 2$, the local equations give

$$Av_i = \frac{(1 + \beta_i)v_i - \beta_{i-1}v_{i-1} - v_{i+1}}{\alpha_i} + \frac{f_i + h_i - h_{i-1}}{\alpha_i}, \quad (40)$$

with negative-index terms zero. The coefficient sum is at most $8 + 256H \leq D$, and the error norm is at most ωD^{i+3} . Inductively,

$$A^t b = \sum_{i=0}^t c_{ti} v_i + e_t, \quad \sum_i |c_{ti}| \leq D^t, \quad \|e_t\|_2 \leq 2\omega D^{2t+1}.$$

The error induction uses $\|A\|_2 = 1$: $\|e_{t+1}\|_2 \leq 2\omega D^{2t+1} + \omega D^{2t+3} \leq 2\omega D^{2t+3}$. Combining the monomials of π yields (37).

Next, substitute $v_i = (x_{i+1} - x_i - g_i)/\alpha_i$ in the direction representation. Since $x_0 = 0$, the error-free part is in the claimed iterate span. Its additional error is at most $4D^{2k}\omega D^{k+1}$; together with e this is below ωD^{3k+2} .

Finally, expand v_i into earlier residuals as in the bootstrap proof. The residual coefficients have sum at most $2D^i$ and the accumulated local error has norm at most $3\omega D^{i+1}$. Since $k > i$,

$$|r_k^T v_i| \leq 2D^i E_k + 3\omega D^{i+1} R_k \leq \omega D^{9k+10}.$$

□

7 The three termination arguments

Theorem 7.1 (Capped-condition, early-horizon upper bound). *With the definitions (2)–(4), every precision satisfying (28) guarantees a successful produced iterate by step m . Therefore (5) and (6) hold.*

Proof. Assume for contradiction that all produced iterates fail. By Lemma 5.3, every iterate through m exists and the bootstrap estimates hold. At least one of the following three cases applies to the minimum defining m .

Case 1: the dimension horizon $m = n$

Normalize $r_0, \dots, r_n$ to unit vectors. Their Gram matrix has diagonal one and off-diagonal entries of magnitude at most

$$\frac{E_n}{\sigma^2} \leq D^{-4n-12}.$$

Every off-diagonal row sum is smaller than one. The inequality $2|a_i a_j| \leq a_i^2 + a_j^2$ shows directly that this Gram matrix is positive definite. Its rank would be $n + 1$, impossible for vectors in $\mathbb{R}^n$.

Case 2: the Chebyshev horizon $m = k_C$

Put $k = m$, $a = 1/\bar{K}$, and

$$q(t) = \frac{T_k((1+a-2t)/(1-a))}{T_k((1+a)/(1-a))}, \quad \pi(t) = \frac{1-q(t)}{t}.$$

Since $q(0) = 1$, π is a polynomial of degree less than k . The affine argument of T_k has coefficient sum at most seven because $a \leq 1/2$. The Chebyshev recurrence gives

$$\|T_k((1+a-2t)/(1-a))\|_{\text{coef},1} \leq 15^k.$$

Its normalizing denominator is at least one. Thus

$$\|\pi\|_{\text{coef},1} \leq 16^k \leq D^{k+1}.$$

On the spectrum of A ,

$$|q(t)| \leq 2\vartheta^k, \quad \vartheta = \frac{\sqrt{K} - 1}{\sqrt{K} + 1}.$$

For $x_* = A^{-1}b$ and $y = \pi(A)b$, it follows that

$$\|x_* - y\|_A^2 \leq 4K\vartheta^{2k}. \quad (41)$$

We now compare y with the actual rounded x_k , not an exact CG projection. Let $\bar{x}_k = \sum_{i \leq k} \alpha_i v_i$, so $\|x_k - \bar{x}_k\|_2 \leq 2\omega D^{k+1}$. By Lemma 6.1, the difference between the direction representation of y and $\bar{x}_k$ has coefficient sum at most D^{2k+1} , and the remaining vector error is at most ωD^{3k+2} . Combining (39) with the residual gap therefore gives

$$\begin{aligned} |(b - Ax_k)^T(y - x_k)| &\leq \omega D^{11k+11} + \omega D^{4k+3} + 8\omega D^{2k+3} \\ &\leq \omega D^{11k+12}. \end{aligned} \quad (42)$$

For the last gap term, we used $\|y\|_2 \leq \|\pi\|_{\text{coef},1} \leq D^{k+1}$ and $X_k \leq D^{k+1}$; this bound does not replace the capped H by K .

The exact energy identity implies

$$\|x_* - x_k\|_A^2 \leq \|x_* - y\|_A^2 + 2|(b - Ax_k)^T(y - x_k)|.$$

Since $A^2 \preceq A$, (41) and (42) yield

$$\|b - Ax_k\|_2^2 \leq 4K\vartheta^{2k} + \omega D^{12k+20}.$$

The chosen k_C makes the first term at most $\varepsilon^2/4$: use

$$\vartheta \leq \exp(-2/\sqrt{K}), \quad k_C \geq \sqrt{K} \log_2(16\bar{K}/\varepsilon).$$

The second term is at most $\varepsilon^2/16$. Thus $\|b - Ax_k\|_2 < \varepsilon$, which implies (1) because its denominator is at least one. This contradicts failure.

Case 3: the universal backward horizon $m = k_U$

Put $k = m$. We use the following exact polynomial fact. For the even integer $\ell \in \{k, k+1\}$, define

$$G(t) = \frac{1 - T_\ell(2t - 1)}{2\ell^2}, \quad \pi(t) = \frac{t - G(t)}{t^2}. \quad (43)$$

The numerator has a double zero at zero; π has degree at most $\ell - 2 < k$, is positive on $[0, 1]$, and satisfies

$$\left| t - \frac{1}{\pi(t)} \right| \leq \frac{3}{k^2 - 1}, \quad 0 \leq t \leq 1. \quad (44)$$

These are Lemmas 12–13 of Dereziński, Nakatsukasa, and Rebrova[2]. This is the only external approximation lemma used in this case. The shifted Chebyshev recurrence gives coefficient sum at most 7^ℓ , so

$$\|\pi\|_{\text{coef},1} \leq 16^{k+1} \leq D^{k+1}.$$

Also (44) gives $1/\pi(t) \leq 2$, so $\pi(t) \geq 1/2$. Therefore $y = \pi(A)b$ satisfies

$$\|y\|_2 \geq \frac{1}{2}, \quad \text{berr}(y) := \frac{\|b - Ay\|_2}{\|y\|_2} \leq \frac{3}{k^2 - 1}. \quad (45)$$

The spectral implication is immediate by applying $A - \pi(A)^{-1}$ to y .

The failure assumption imposes an incompatible lower bound on every vector in the computed-iterate span. Indeed, for $z = \sum_{j=1}^k c_j x_j$,

$$b - Az = \left(1 - \sum_{j=1}^k c_j\right) r_0 + \sum_{j=1}^k c_j r_j + \sum_{j=1}^k c_j w_j. \quad (46)$$

The Gram matrix of normalized $r_0, \dots, r_k$ has off-diagonal entries at most

$$\frac{64E_k}{49\varepsilon^2} \leq \frac{4}{49} D^{-4k-12},$$

so its off-diagonal row sums are below $1/4$. In addition, (31) and (32) give $W_j/R_j \leq 1/(8\sqrt{k})$. Write $Z = (\sum_{j=1}^k c_j^2 R_j^2)^{1/2}$. The first two terms in (46) have norm at least $\sqrt{3/4} Z$, while the last has norm at most $Z/8$. Thus $\|b - Az\|_2 \geq Z/2$. The failure bound $X_j/R_j \leq 2/\varepsilon$ and Cauchy–Schwarz give $\|z\|_2 \leq 2\sqrt{k} Z/\varepsilon$. Every nonzero vector in this span therefore satisfies

$$\text{berr}(z) \geq \frac{\varepsilon}{4\sqrt{k}}. \quad (47)$$

Lemma 6.1 supplies a vector in the same span with

$$\|z - y\|_2 \leq \omega D^{3k+2} \leq \frac{\varepsilon}{128\sqrt{k}}.$$

Relative to $\|y\|_2 \geq 1/2$, this is at most $t_0 = \varepsilon/(64\sqrt{k})$, so $z \neq 0$. The definition of k_U gives

$$\frac{3}{k^2 - 1} \leq \frac{4}{k^2} \leq \frac{\varepsilon}{16\sqrt{k}}.$$

Perturbing (45) now gives

$$\text{berr}(z) \leq \frac{\text{berr}(y) + t_0}{1 - t_0} \leq \frac{\varepsilon}{8\sqrt{k}},$$

contradicting (47).

All possible minimizers of the horizon lead to a contradiction. Thus a successful iterate is produced by m . Condition (28) is exactly implied by

$$u \leq \frac{\varepsilon^2}{1024(n+1)KD^{12m+20}}.$$

Every larger precision satisfies the same inequality, and the argument applies to all real normalized inputs. This proves the full threshold quantifier, (5), and its order form (6). $\square$

8 Consequences with matching bounds

Corollary 8.1 (Bounded conditioning). *Uniformly for $n \geq 1$, $1 \leq K \leq 2$, and $0 < \varepsilon < 1/2$,*

$$P_{\text{CG}}(n, K, \varepsilon) = \Theta\left(\log_2 \frac{n}{\varepsilon}\right).$$

Proof. Here $H \leq 2$ and $D \leq 4096$ are universally bounded. Moreover $m \leq k_C = 2\lceil \log_2(32/\varepsilon) \rceil$. Equation (5) is therefore $O(\log(n/\varepsilon))$. The lower bound is (15). $\square$

Corollary 8.2 (Additive characterization at fixed tolerance). *For every fixed $0 < \varepsilon < 1/2$, uniformly over all $n \geq 2$ and $K \geq 1$,*

$$P_{\text{CG}}(n, K, \varepsilon) = \log_2(nK) + O_\varepsilon(1).$$

An explicit lower additive constant is -5 . An upper additive constant may be taken as

$$C_\varepsilon = 1 + \log_2 \frac{1536}{\varepsilon^2} + (12k_U + 20) \log_2 \left(1024(64/\varepsilon^2)^2 \right).$$

Proof. The lower inequality is (18). For the upper inequality, use $H \leq 64/\varepsilon^2$, $m \leq k_U$, and $n+1 \leq 3n/2$ in (5), and allow one bit for its ceiling. Every term remaining after $\log_2(nK)$ depends only on ε . $\square$

Corollary 8.3 (Leading conditioning coefficient). *For every fixed $n \geq 2$ and fixed $0 < \varepsilon < 1/2$,*

$$P_{\text{CG}}(n, K, \varepsilon) = \log_2 K + O_{n,\varepsilon}(1), \quad \lim_{K \rightarrow \infty} \frac{P_{\text{CG}}(n, K, \varepsilon)}{\log_2 K} = 1.$$

Proof. Once $K \geq 64/\varepsilon^2$, the quantity $H = 64/\varepsilon^2$ no longer depends on K . Since $m \leq n$, (5) is at most $\log_2 K + C_{n,\varepsilon}$ for a finite constant independent of K . On the other hand, (16) is at least $\log_2 K - 1$. These two inequalities prove both assertions. $\square$

This fixed-tolerance limit is compatible with the recovered $n \log K$ lower family: that family decreases ε as K increases. Interchanging these parameter regimes would give an incorrect conclusion.

Corollary 8.4 (Two dimensions and very fine tolerances). *Equation (10) holds uniformly. More generally, if $n \geq 2$ and $\log_2(1/\varepsilon) \geq n \log_2(2K)$, then*

$$P_{\text{CG}}(n, K, \varepsilon) = \Theta\left(\log_2 \frac{nK}{\varepsilon}\right)$$

with universal constants on this region.

Proof. Use $H \leq K$ and $m \leq n$ in (5). When $n = 2$, its logarithm is $O(\log K + \log(1/\varepsilon))$. Under the second displayed condition, the term $n \log(2K)$ is absorbed by $\log(1/\varepsilon)$. Theorem 3.1 supplies the matching lower bounds. $\square$

At $K = 1$, the tighter first-step estimate from the recovered manuscript is $P_{\text{CG}} \leq \lceil \log_2(128(n+1)/\varepsilon) \rceil$. Together with (15), this yields the additive form $P_{\text{CG}}(n, 1, \varepsilon) = \log_2(n/\varepsilon) + O(1)$ with universal additive constants. The exact scalar result is stronger still when $n = 1$.

9 The retained hard family and the unresolved gap

The recovered manuscript proves the following separate lower-bound family. Let $s = n - 1$, and let M_s have diagonal $(1, 2, \dots, 2)$ and adjacent off-diagonal entries -1 . Its inverse bidiagonal factor gives $\lambda_{\min}(M_s) \geq 1/s^2$, and $\|M_s\|_2 < 4$. Take

$$T = 2^\ell, \quad A = M_s \oplus [T], \quad b = (e_1, 1)^T, \quad p = s\ell.$$

The only nonzero operation error is $+2^{-p}$ on the last addition in ρ_0 . All later operations are exact. In particular the true and recursive residuals agree. For each fixed $n \geq 2$, the recovered recurrence analysis shows that, for all sufficiently large integers ℓ ,

$$P_{\text{CG}}\left(n, s^2 2^\ell, \frac{1}{64s^2 2^\ell}\right) > s\ell. \quad (48)$$

The quantifier is fixed n , followed by sufficiently large ℓ ; no dimension-uniform threshold is supplied. The full proof remains in Section 5 of the preserved prior manuscript. Its finite exact witnesses have been rerun in this package. The new results do not strengthen the quantifier in (48).

A concrete gap still present. Consider $n \geq 2$ with

$$K = n^2, \quad \varepsilon = n^{-2}. \quad (49)$$

The generally applicable lower bound established in this package is $\Omega(\log n)$, while its upper bound is $O(n \log n)$. The retained large-outlier limit does not close this growing-dimension regime. Neither an $\Omega(n \log n)$ lower bound nor an $O(\log n)$ upper bound on (49) is proved here.

A full solution must supply a universally matching characterization on such regimes as well as on the sharp regions already covered. The upper bound in this manuscript is not asserted to be sharp everywhere. In particular, its reciprocal-polynomial horizon is a sufficient precision budget, not a matching lower bound when n exceeds that horizon. An interval-polynomial obstruction is also not, by itself, an admissible CG execution; the actual recurrence and its spectral weights must be checked.

The reason for the partial label is this specific mathematical gap, not merely the problem’s status in a repository. No full-resolution or priority claim is made.

10 Exact checks, reproducibility, and proof scope

All executable checks use Python’s `fractions.Fraction` and the standard library. No approximate square root or floating logarithm is used to decide a certificate. The error-envelope executor checks input significands, symmetry, exact LDL^T positivity, and every supplied operation-error magnitude. It includes the required zero terms and separates products from additions.

10.1 What was checked

The regenerated reports record:

Check	Count
Scalar first-step sign-corner executions (three precisions)	1,536
First-step executions at a certified precision (finite inputs/schedules)	256
Exact reciprocal-polynomial constructions	31
Rational-grid reciprocal-polynomial inequalities	3,999
Parameter-triple consistency checks for bound evaluation	90
Joint dimension/conditioning breakdown witnesses	14
Dense matvec prefix-bound executions	24
Retained exact counterexample witnesses	15
Unit tests, including the retained regression suite	26

The first-step executions use diagonal SPD inputs with known exact spectral norm, dimensions 2, 3, 5, 8, condition bounds 1, 2, 2^6 , 2^{20} , two coarse tolerances, and eight deterministic error schedules. These are finite checks, not an exhaustive multivariate error search. The scalar corner enumeration is exhaustive only for its nine corner signs; its continuous-box validity is proved analytically in Theorem 2.1.

For known rational $a = \|A\|_2$ and squared norms R^2, X^2, B^2 , testing $\eta \leq \varepsilon$ reduces to comparing

$$C = R^2/\varepsilon^2 - a^2X^2 - B^2 \quad \text{with} \quad 2a\sqrt{X^2B^2}.$$

If $C < 0$, success is strict. If $C \geq 0$, compare the squares, retaining the zero case separately. This gives exact comparison without extracting a square root. The older hard-family verifier uses an independent sufficient failure certificate based on $(aX + B)^2 \leq 2(a^2X^2 + B^2)$.

10.2 Commands and bound evaluation

From the package root, run

```
python -m unittest discover -s code -v
python code/verify_extended.py
python code/verify_legacy.py --output results/legacy_checks.json
python code/bounds.py --n 16 --K 256 --epsilon 1/256
```

Run without `-O`; the witness scripts reject optimized mode because they use assertions. Python 3.10 or later is sufficient.

The scalar evaluator returns the exact integer threshold. The general evaluator reports separate lower and upper bounds, not an alleged exact answer. To avoid constructing enormous rational powers, its conservative integer version of (5) is

$$\left\lceil \log_2 \frac{1024(n+1)K}{\varepsilon^2} \right\rceil + (12m+20)\lceil \log_2 D \rceil.$$

This can exceed the ceiling in the theorem by at most $12m+20$ bits. It remains a valid sufficient bound and has the same stated order. The one-step evaluator compares the square of $\varepsilon - 512(n+1)K2^{-p}$ with $M(K)^2$, after checking the sign.

10.3 What the checks do not establish

Passing these checks does not prove the bootstrap for arbitrary inputs, the uniform reciprocal-polynomial inequality between grid points, the large-outlier asymptotics in arbitrary dimension, or a matching precision formula. The bootstrap, transfer arguments, and corollaries are written mathematical proofs; the reciprocal-polynomial inequality has a cited analytic proof. None is represented as formally verified or independently reviewed. The preserved old archive is also explicitly partial.

A A compact proof-dependency audit

The scalar theorem depends only on counting the specified nine operations and monotonicity of a rational expression. The first-step theorem depends on the moment inequality and local scalar envelopes. The main upper bound uses the following chain, with no circular assumption of successful termination:

Stage	Required fact
Local equations	Actual K controls denominator perturbations.
Current step legality	Earlier defect bounds give $r_j^T v_j > 0$.
Capped step length	Failure of the newly produced iterate rules out a long step.
Bootstrap propagation	Coefficient bounds, local errors, and adjacent cancellations.
Horizon contradiction	Rank, Chebyshev energy comparison, or reciprocal-polynomial span comparison.
Threshold conclusion	The same sufficient inequality holds at every larger precision.

The joint breakdown lower bound is separate from this chain: it uses exactly representable inputs and rational, greedy accumulation factors.

The only non-elementary external approximation fact used in the new upper proof is (44). Removing its branch leaves the still-valid upper bound with $m = \min\{n, k_C\}$, including the cap, the bounded- K order, and the leading large- K coefficient.

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